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SUPERSEDING
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MILITARY SPECIFICATION

ENGINE, DIESEL: 6-CYLINDER, V-TYPE,
300 H.P. TURBO-CHARGED

1. SCOPE

1.1 Scope. This specification covers one type of commercial 6-cylinder, V-type, liquid-cooled, 2-stroke-cycle, turbo-charged internal-combustion, compression ignition engine for use in military vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-I-490	-	Inhibitor Corrosion, Liquid Cooling System.
P-C-437	-	Cleaning Compound, High Pressure (Steam) Cleaner.
VV-F-800	-	Fuel Oil, Diesel.

Military

MIL-P-514	-	Plates, Identification, Transportation Data and Blank.
MIL-L-2104	-	Lubricating Oil, Internal-Combustion Engine, Heavy-Duty.
MIL-C-11755	-	Compound, Antifreeze, Arctic-Type.
MIL-S-13856	-	Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for.
MIL-L-21260	-	Lubricating Oil, Internal-Combustion Engine, Preservative and Break-In.
MIL-A-46153	-	Antifreeze, Ethylene Glycol, Inhibited.
MIL-L-46167	-	Lubricating Oil, Internal-Combustion Engine, Sub-Zero.

FSC-2815

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STANDARDS
Federal
FED. TEST METHOD STD. NO. 791 - Lubricants, Liquid Fuels, and Related Products· Method of Testing.

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

DRAWING
ORD-11669078 - Engine, Diesel (Detroit Diesel Engine No. 5103048).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 First article test. The engine shall be subject to preproduction or initial production tests, as applicable, and as invoked in contract (see 4.2).

3.2 Construction. Construction and assembly shall conform to Ord. Drawing 11669078.

3.2.1 Accessories and equipment. All electrical accessories and equipment, except the engine starter and air pump wiring harness, shall conform to the applicable requirements of Specification MIL-E-13856. Unless otherwise specified, all accessories and equipment shall be installed on the engine and properly adjusted.

3.3 Interchangeability of parts. Component assemblies and parts of the engine shall be so constructed that any part, except those furnished in matched sets or for which a selective fit is specified, may be installed, replaced, and adjusted without requiring modification.

3.4 Operational requirements.

3.4.1 Environmental requirements. The engine shall start within 2 minutes and shall demonstrate the performance characteristics specified herein under any of the following conditions or possible combination of conditions:

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3.4.1.1 Temperature conditions.

(a) With integral cold starting aid, and without external aids after being cold-soaked for a 24 hour period, without benefit of solar radiation, to an ambient temperature of -25°F. The engine cranking speed should not be less than 120 RPM and the voltage to the air pump should not be less than 16 volts.

(b) Without external aids, with cooling fan installed, when operated at the following temperatures with temperatures and exposure to maximum solar radiation:

Ambient air temperature	125°F.
Intake air temperature (at air cleaner inlet)	125°F.
Coolant outlet temperature	230°F.
Oil sump temperature	At maximum temperature attainable up to 275°F.

3.4.1.2 Elevation conditions. At any elevation from sea level to 8,000 feet, except that the power requirements of the engine shall be reduced proportional to the elevation, so that at 8,000 feet elevation, the observed power output of the engine shall be not less than 90 percent of rated power.

3.4.1.3 Humidity conditions. Under relative humidity conditions as low as 5 percent at temperature of 115°F. and as high as 100 percent at all temperatures from -25 to 85°F

3.4.2 Coolant. The engine shall perform as specified herein using water and antifreeze compound conforming to Specification MIL-A-46153, arctic-type antifreeze compound conforming to Specification MIL-C-11755, or water and corrosion inhibitor compound conforming to Specification O-I-490.

3.4.3 Limiting operating temperatures. The engine shall not exceed the following limiting temperatures when operating at full load throughout the speed range of the engine under any of the conditions specified herein:

(a)	Oil sump	275°F.	(measured at the drain plug).
(b)	Coolant	230°F.	(measured at coolant outlet).

3.4.4 Electromagnetic compatibility. Unless otherwise specified, engine shall meet the functional suppression requirements class IIIA of MIL-STD-461 as specified.

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3.4.5 Grades and slopes. The engine shall start and demonstrate the performance characteristics specified herein when operated throughout the speed range of the engine, with full and one quart low oil level conditions in the sump, on longitudinal grades up to 60 percent or lateral slopes up to 40 percent in all directions and, as a result of said operation, no evidence of faulty lubrication, cooling, fuel supply, leakage, or other malfunction shall be found.

3.4.6 Steam and water jet cleaning. The engine and all its components shall withstand cleaning with high pressure steam and cleaner conforming to Specification P-C-437A and water jet, without deterioration of seals or hose, water past seals or gaskets, or other defects. Paint removal shall not be considered a deficiency. Steam and water jet cleaning is defined as follows.

> The jet is applied perpendicular to the surface being cleaned at a distance of not more than 1 ft. from the surface for steam cleaning, and not more than 5 ft. from the surface for water jet cleaning, and cleaned at a rate of 1 square foot per minute. The jet pressure shall be not more than 110 psi.

3.4.7 Air cleaner. The engine shall perform as specified herein when using a dry type air cleaner or approved design and capacity with a restriction equivalent to 20 inches of water.

3.4.8 Fuel. The engine shall perform as specified herein using fuel in accordance with Specification VV-F-800.

3.4.8.1 For production break-in run. During the production break-in run, fuel conforming to Grade 2 of Specification VV-F-800, shall be used (see 3.6).

3.4.8.2 For four hundred-hour NATO test. Fuel to be used during the 400-hour endurance test shall be as follows:

> Engine shall be operated with diesel fuel conforming to the following: Using method No. 340-.3 of FED. TEST METHOD STD. NO. 791, referee grade diesel fuel shall have the following properties:

Flash point, minimum, °F.	100 or legal
Pour point, maximum, °F.	20
Water and sediment, by volume, maximum, percent	.05
Distillation, 50% evaporated, minimum, °F.	500
90% evaporated, °F.	600-640
End point, °F.	650-690
Kinematic viscosity, centistokes, at 100°F.	1.6 to 4.5
*Sulfur percent	.95-1.05
Corrosion	pass
Alkali and mineral acids	none
Cetane number	40 to 45

*The sulfur must be natural sulfur (See note 1)

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1/ The use of additives of any kind to produce a clean fuel, or solvent-treating the fuel, is not permitted.

3.4.8.3 For 50-hour control test. Fuel conforming to Grade 2 of Specification VV-F-800 shall be used during the 10-hour quality control test (see 4.1).

3.4.9 Oil. Engine shall perform as specified herein using lubricating oil in accordance with the seasonal requirements of Specification MIL-L-2104 from -10 to +115°F., and Specification MIL-L-46167 from -65 to 0°F. The manufacturer may use oil of his own selection during engine build-up and break-in run. Oil conforming to Grade 2 of Specification MIL-L-21260 may be used during engine build-up and production runs specified in 3.6.

3.5 Performance requirements.

3.5.1 Speed range. The engine shall operate satisfactorily under all loads and conditions as specified herein throughout the entire speed range. The engine shall maintain a satisfactory idle speed of 550 to 600 RPM when using diesel fuel conforming to Grade 2 of Specification VV-F-800 (see 3.4.8).

3.5.2 Governor. The governor shall limit the maximum engine speed as follows:

Full load operation	2775 to 2825 RPM
No load operation	2915 to 2965 RPM

The engine speed with no load on the engine shall stabilize within 3 seconds after full rack position is reached.

3.5.3 Brake horsepower. At the completion of the production break-in run (see 3.6 and 4.5.2) the bare engine (see 6.3), at the full rack setting shall develop 282 minimum gross brake horsepower at 2800 RPM when using diesel fuel as specified in 3.4.8. Air density of 0.070 pounds per cubic foot, a fuel temperature of 90°F and fuel specific gravity of 0.853 at 60°F with maximum fuel flows of 130 pounds per hour at 2800 RPM and 115 pounds per hour at 2400 RPM.

3.5.3.1 Correction factors. Engine rating can be corrected as follows: 2.1 horsepower per each 10°F over 85°F air inlet temperature. .72 brake horsepower per each .001 change of fuel specific gravity. One (1) brake horsepower per each 4°F change of fuel temperature.

3.5.4 Torque. At the completion of the production break-in run and with the full rack setting specified in 3.5.3, the bare engine (see 6.3) shall develop at the engine flywheel a minimum observed torque of 566 lb. ft. at 2200 RPM when using fuel as specified in 3.4.8.

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3.5.5 Exhaust smoke density. The maximum exhaust smoke density at full rack position, through the speed range of 2000 to 2800 RPM when measured within three feet of the turbocharger outlet, shall not exceed a No. 3 (light gray) smoke condition as defined in 6.6 when using fuels in accordance with 3.4.8.

3.5.6 Fuel consumption. The fuel consumption of the bare engine (see 6.3) shall not exceed the following limits when operating at the full rack setting specified in 3.5.3, and with fuel as specified in 3.4.8.

Maximum observed brake specific fuel consumption

RPM	Lbs. Per BHP-HR
2200	0.450
2800	0.440

3.5.7 Oil consumption. After 25 hours of operation, the engine shall consume not more than 0.0035 lbs. of lubricating oil per observed brake horsepower hour, when operating at full rack throughout the power check phase (see 4.5.7) of the engine with oil sump temperatures of 200 to 275°F., using oil as specified in 3.4.9.

3.5.8 Oil pressure. Under all operating conditions specified herein, including a one quart low sump condition, with an oil sump temperature range from 200 to 275°F., gallery oil pressure shall not be more than 80 pounds per square inch (psi) and not less than 32 psi, when the engine is operating at 2800 RPM, using oil as specified in 3.4.9, and shall not be less than 5 psi when engine is idling as specified in 3.5.1, with oil sump temperature range from 180 to 260°F.

3.5.9 Turbocharging. The turbocharger shall develop a minimum compressor discharge pressure of 50 in. Hg Absolute when operating at full load and 2800 RPM, using fuel as specified in 3.4.8 and correction factor specified in 3.5.3.1.

3.5.10 Endurance. The engine shall operate as specified in 4.2.4.7 for a period not to exceed 400 hours on the fuels specified in 3.4.8.2. During the test, the engine shall require only minor repairs, such as replacing belts and filters. At the conclusion of this test, the engine shall conform to the operational and performance requirements specified in 3.4 and 3.5, with the exception that the engine shall develop not less than 90 percent of its initial power output with a maximum exhaust smoke density as specified in 3.5.5 without adjustment to the engine and shall develop 95% of full rated power after authorized tuneup procedure. In addition, there shall be no evidence of abnormal wear such as scuffing or galling of cylinder bores, pistons, piston rings, piston pins, bearings, bearing journals, gear teeth, cam surfaces, valves, tappets or other parts, burning of valves or pistons, sticking of valves or piston rings, leakage of gaskets or seals, imminent failure of a component or other malfunction.

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3.6 Production break-in run. A production line run-in shall be performed by the contractor to insure that the engine, as offered for delivery, may be subjected to any or all checks or tests (complete and in sequence) of performance and endurance as set forth in this specification with no loss in life expectancy as stated in 3.5.10.

3.7 Log sheet. The contractor shall maintain a log sheet of each engine run during all tests specified herein. The form of the log sheet and number of legible copies to be furnished to the Government shall be as approved by the procuring activity.

3.8 Exterior surface treatment. All exposed exterior surfaces of the engine and its components shall be cleaned, painted, or treated for corrosion resistance as specified on the applicable drawings, or if not so specified, in accordance with applicable provisions of Standard MIL-STD-193.

3.9 Product identification.

3.9.1 Nameplate. Unless otherwise shown on the applicable drawings, the nameplate, data, and instruction plates shall conform to Specification MIL-P-514.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's inspection.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Contractor's quality control system. The supplier shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer no later than 30 days prior to initiation of production for determination of acceptability. The supplier will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The supplier shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

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4.1.3 Government verification. All quality assurance operations performed by the supplier will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices, which might have an affect upon the quality of the product, will be immediately called to the attention of the supplier. Failure of the supplier to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 First article test.

4.2.1 First article testing. The tests apply to preproduction or initial production engines and will be required when cited in contract.

4.2.1.2 Test. The engine selected for test in accordance with 4.2.1 shall have break-in (4.3.1), acceptance (4.3.2) and examination (4.3.3) prior to test specified in Table I.

Table I - First article testing

1.	Starting Tests	See 4.2.2
2.	Slope	See 4.2.3
3.	400 Hour Endurance	See 4.2.4
4.	Power Check	See 4.2.5
5.	Elevation	See 4.2.6
6.	High Temperature	See 4.2.7
7.	Steam & Water Cleaning	See 4.2.8
8.	Electromagnetic Compatibility	See 4.2.9
9.	Teardown	See 4.2.10

4.2.2 Starting tests.

4.2.2.1 Cold starting. To determine conformance to item a of 3.4.1.1, the bare engine, with integral winterization aids, shall be cold started at -25°F on fuel specified in 3.4.8. Prior to the cold start, the engine shall be cold soaked in an ambient temperature of -25°F for a period of 24 hours. After each start, the engine must operate for a sufficient time to stabilize engine operation.

4.2.2.2 High temperature starting. The bare engine, without aids, shall be placed in a high temperature chamber maintained at a temperature of 125°F and operated at full load until all temperatures are stabilized within the limits

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specified in 3.4.3. The engine shall then be shut down, hot soaked, and restarted when the fuel temperature at the inlet to the injection pump reaches its maximum value. Fuel shall be supplied to the engine mounted fuel supply pump at a pressure of 5 psi. The fuel pressure shall be shut off during the hot soak period. The starting time shall not exceed 2 minutes, to determine conformance to item b of 3.4.1.1.

4.2.3 Slope test. The engine shall be tested to determine conformance to 3.4.5.

4.2.4 Endurance test. The bare engine (see 6.3) shall be operated in accordance with Table II to determine conformance to 3.4.10.

4.2.4.1 Data. The following data shall be recorded at the end of each hour of operation, at the completion of each period, and just prior to stopping the engine (emergency stops excluded):

a. Engine Speed, RPM
b. Observed Brake Horsepower (BHP)
c. Observed Torque (Lbs-Ft)
d. Air Box Pressure (In Hg)
e. Exhaust Manifold Pressure, In Hg (Before Turbocharger)
f. Fuel Pressure After Secondary Filter, Psi
g. Fuel Supply Pressure at Transfer Pump, Psi
h. Lubrication Oil Pressure (Gallery), Psi
i. Crankcase Pressure, Inches of Water
j. Lubricating Oil Temperature, ^{o}F (Sump)
k. Lubricating Oil Temperature Inlet, ^{o}F (Gallery)
l. Fuel Temperature After Secondary Filter, ^{o}F
m. Air Temperature Into Turbocharger, ^{o}F
n. Air Temperature at Air Box, ^{o}F
o. Coolant Temperature, Engine Outlet
p. Exhaust Temperature After Turbocharger
q. Test Cell Ambient Air Temperature
r. Blowby, CFM
s. Fuel Flow, Lbs/Hr
t. Specific Fuel Consumption, Lbs/Bhp-Hr
u. Specific Oil Consumption, Lbs/Bhp-Hr (at Power Check)
v. Barometric Pressure of Test Cell, Once Each 4-Hour Period
w. Exhaust Smoke Density (see 6.6)

4.2.4.2 Test hours. Running time less than 30 minutes unless specified in Table II shall not be counted toward fulfillment of endurance hours.

4.2.4.3 Fuel. During this test, the engine shall be operated on the fuel specified in 3.4.8.2.

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4.2.4.4 Oil. During this test, the engine shall be operated on referee grade oil (see 3.4.9 and 6.4).

4.2.4.5 Temperatures. Inlet air temperatures shall be not less than 70°F nor more than 100°F unless otherwise specified (see 4.2.7). The lubricating oil temperature shall not exceed 275°F.

4.2.4.6 Servicing. Prior to the power check test, the lubricating oil shall be changed, all filters serviced, valve clearances, and injector rack settings checked and adjusted to applicable requirements.

4.2.4.7 Acceptable endurance performance. During the endurance test, the engine shall be observed for compliance with the following performance requirements during the power check tests (see 4.2.5). The power checks shall be made at the beginning of the endurance run and at the end of each 100 hour period.

a.	Speed Range	See 3.5.1
b.	Governed Speeds	See 3.5.2
c.	Brake Horsepower	See 3.5.3
d.	Torque	See 3.5.4
e.	Exhaust Smoke Density	See 3.5.5
f.	Fuel Consumption	See 3.5.6
g.	Oil Consumption	See 3.5.7
h.	Oil Pressure	See 3.5.8
i.	Supercharging	See 3.5.9
j.	Engine Life	See 3.5.10

Table II - Endurance Run (NATO Cycle)

The endurance test shall be 400 hours, consisting of four periods of 100 hours each. Each 100-hour period is to be made of 20 5-hour schedules. Each 5-hour schedule will be in accordance with the following program:

Period	% Rated Speed	% Load	Duration Hours
1	Idle	0	1/2
2	Max Torque	100	1
3	100	0	1/2
4	75	85	1
5	100	50	1/2
6	100	100	1
7	50	25	1/2
		TOTAL DURATION	5

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4.2.5 Power check test. Full rack performance characteristics of the engine shall be determined by operating the engine according to the following schedule;

Period	Engine Speed	Rack Position	Fuel
1	1800	Full Rack	See 3.4.8.2
2	2000	Full Rack	See 3.4.8.2
3.	2400	Full Rack	See 3.4.8.2
4	2600	Full Rack	See 3.4.8.2
5	2800	Full Rack	See 3.4.8.2

All data necessary to determine the performance characteristics specified in 4.2.4.7 or 4.4.1.1 as applicable, shall be recorded in the log. Performance curves showing horsepower, torque and specific fuel and oil consumption shall be required in addition to the log sheets. Oil consumption data shall be obtained from endurance hours. Operating conditions (3.5.3.1) shall be maintained during test.

4.2.6 Elevation test. The engine shall be placed in a simulated environmental condition of 8,000 feet elevation, and its general performance characteristics observed to determine conformance to 3.4.1.3.

Eleveation Test	Pressure Inches of Mercury*	Temperature °F
8,000	22.2	90

*NOTE: Measure at turbocharger inlet and exhaust outlet.

4.2.7 High temperature test. To determine compliance with item c of 3.4.1.1, the last 100 hours of the 400-hour endurance test shall be conducted with the intake air temperature of 115° ∠ 5°F at an ambient air temperature of 115° ∠ 5°F at the fuel transfer pump inlet. The power check tests shall be made at standard operating conditions (see 3.5.3.1).

4.2.8 Steam and water jet cleaning test. At the 400-hour endurance test (see 4.2.4), and just prior to the tear down inspection, the engine shall be subjected to normal steam cleaning with cleaner as specified in 3.4.6. Immediately upon completion of the steam cleaning operation, the engine shall be subjected to normal water jet cleaning as specified in 3.4.6. Prior to cleaning, the air intake opening, flywheel, exhaust, and all other openings that might be affected shall be covered. The engine shall start and operate at idle speed without misfiring within two minutes immediately after the completion of both the steam and water jet operations. During the disassembly of the engine, special notice should be made of water entering the engine through joints, seals, gaskets, etc.

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4.2.8.1 Engine assembly systems malfunction. Evidance of malfunction or of water seepage past seals, gaskets, joints, etc., shall constitute a failure of the test.

4.2 9 Electromagnetic compatibility test. To determine conformance to 3.4.4, engine shall be tested for meeting electromagnetic compatibility requirements specified (see 6.7).

4.2.10 Teardown inspection. Final inspection after first article test will include teardown of the engine for detailed investigation.

4.3 Production tests.

4.3.1 Production break-in run. Each engine shall be operated in accordance with 3.6, using fuel specified in 3.4.8.1. Any unsatisfactory operation of conditions which require correction shall be corrected by the contractor before the engine is presented for acceptance.

4.3.1.1 Teardown inspection. At the conclusion of the break-in run, the supplier in the presence of the Government inspector shall disassemble and inspect the first 10 engines, and every second engine of the next 10 of any contract except overlapping contracts. Disassembly shall be accomplished to the extent required to perform the following inspection:

a. Oil Contamination
b. Dirt, Chips or Foreign Matter in the Engine Block, Oil Pan, Filters, and Accessory Drive Case
c. Main Bearing Bolt Torque
d, Connecting Rod Bolt Torque
e. Cylinder Bore Scuffing, Scoring, Galling, etc.
f. Piston Scuffing and Burning

4.3.1.2 Corrections and reassembly. Should there be evidence of any condition of engine or parts requiring correction, such corrections shall be made by the contractor prior to reassembly of the engine and resubmission to the break-in run. In addition, the contractor shall submit objective evidence to the Government that corrective action has been taken in production to eliminate any deficiency disclosed. Failure of the contractor to submit such evidence shall be cause for refusal by the Government to continue acceptance of subsequent engines. Any engine requiring replacement parts shall be reassembled and subjected to the break-in run (4.3.1).

4.3.2 Acceptance tests. Engines submitted to the Government for acceptance shall be observed for compliance with the following performance requirements upon completion of the break-in run:

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a.	Speed Range	See 3.5.1
b.	Governor Setting	See 3.5.2
c.	Brake Horsepower	See 3.5.3
d.	Torque	See 3.5.4
e.	Exhaust Smoke Density	See 3.5.5
f.	Oil Pressure	See 3.5.8

4.3.3 Examination. Each of the engines selected in accordance with 4.4.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table IV.

4.4 Control tests.

4.4.1 Tests. A 50-hour control test shall be conducted as specified herein for the purpose of determining the continued quality of the engine. The test is required at the rate of one per month when production is 100 units per month or less, and at a rate of two per month when production is greater than 100 units per month. The engine shall be identified as to production period and shall have completed break-in (4.3.1), acceptance (4.3.2) and examination (4.3.3) prior to the control test specified in Table III.

Table III - 50-Hour Control Test

Period	Time (Hrs)	Speed (RPM)	Rack Position (% of Full Rack)	Fuel
1	(Run-In Schedule)			Fuel conforming to VV-F-800 shall be used during the entire run
2	5	1800	50	
3	5	2300	75	
4	5	2800	75	
5	5	2500	100	
6	30	2800	100	
7	Power Check Test (see 4.2.5)			

4.4.1.1 Performance. During the 50-hour control test, the engine shall be observed for conformance to the following performance requirements:

a.	Limiting Operating Temperatures	See 3.4.3
b.	Speed Range	See 3.5.1
c.	Governor Setting	See 3.5.2
d.	Brake Horsepower	See 3.5.3

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e.	Torque	See 3.5.4
f.	Exhaust Smoke Density	See 3.5.5 and 6.6
g.	Fuel Consumption	See 3.5.6
h.	Oil Consumption	See 3.5.7
i.	Oil Pressure	See 3.5.8
j.	Turbocharging	See 3.5.9

4.4.1.2 Temperatures. The coolant temperature shall be maintained at 180° ± 5°F.

4.4.2 Data. The following recordings are required during control test:

a. Engine Speed, RPM
b. Observed Brake Horsepower, BHP
c. Observed Torque Lb-Ft
d. Air Box Pressure (In Hg)
e. Fuel Pressure After Secondary Filter, Psi
f. Lubricating Oil Pressure, Psi
g. Lubricating Oil Temperature in Oil Sump, °F
h. Fuel Temperature After Secondary Filter, °F
i. Coolant Temperature - Engine Outlet, °F
j. Air Temperature at Turbocharger, °F
k. Cell Ambient Air Temperature, °F
l. Exhaust Temperature (After Turbocharger)
m. Fuel Flow, Lb/Hr
n. Brake Specific Fuel Consumption, Lbs/Obs Bhp-Hr
o. Specific Oil Consumption, Lbs/Obs Bhp-Hr
p. Exhaust Smoke Density
q. Barometer Pressure During Power Check Test

NOTE: All data shall be obtained under stabilized operating conditions, with engine coupled to a dynamometer. At each start, the engine shall be warmed up until torque, speed and temperatures have been stabilized for a period not less than 1 minute. All applicable temperatures shall be recorded at the conclusion of each warm-up period. Warm-up starting and stopping times shall be recorded; however, warm-up time and running time of less than 30 minutes, unless specified in Table III, shall not be counted toward fulfillment of test run hours. Prior to test, all applicable settings and adjustments and setting shall be reset to specifications before test is begun.

4.4.3 Control test failure. Failure of a control test sample to pass any specified examination or test shall be cause for the Government to refuse acceptance of subsequent lots until it has been proven to the satisfaction of the Government that the faults revealed by the test have been corrected.

4.4.4 Teardown. Upon completion of the test specified in 4.4.1, the engine shall be disassembled to the extent stipulated in 4.3.1.1 for inspection. Evidence

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of component or parts damage indicated by this inspection shall require such further disassembly of the engine necessary to determine the cause of failure and correction.

4.4.5 Disposition of engine. After correction of all defects by the test, engine shall be reassembled, tested as specified in 4.3.2, inspected as specified in 4.3.3 and, if found acceptable, shipped as a regular production engine.

4.5 Sampling for lot acceptance.

4.5.1 Inspection lot. A lot shall consist of all engines of one type from an identifiable production period, submitted at one time for acceptance.

4.5.2 Sampling for examination. Samples for examination shall be selected in accordance with inspection level II of Standard MIL-STD-105.

Table IV - Classification of Defects
In Accordance with Standard MIL-STD-105

	Categories	Defects	Gage or Other Method of Inspection
	Critical:	None Defined	
	Major:	AQL 6.5%	
101	Valve Tappet Clearance Each Valve	Intake and Exhaust Improperly Adjusted	Gage
102	Oil Sump, Fuel and Oil Filters	Dirt, Chips and Foreign Particles, Due to Production	Visual and Functional
103	Fuel, Oil and Coolant Leaks	Leakage	Visual
104	Fuel Lines	Damaged	Visual and Functional
105	Governor	Malfunction	Visual and Functional
106	Torque on Cylinder Head Bolts, Intake and Exhaust Manifold Flange Bolts, Vibration Damper	Improper	Torque Wrench
107	Crankshaft	Excessive or Restrictive End Play	Gage and Functional
108	Engine	Malfunction	Functional

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Table IV (Cont) - Classification of Defects
In Accordance with Standard MIL-STD-105

	Categories	Defects	Gage or Other Method of Inspection
109	Fuel System Components	Malfunction, Damage or Leaks, etc.	Visual and Functional
110	Fuel Injection System, Timing and Components	Malfunction, Improper Adjustment	Visual and Functional
	Minor:	AQL 25 per 100 units	
201	Linkage	Improperly Adjusted	Visual
202	Minor Assemblies	Omitted	Visual
203	Engine	Improper Adjustment or Installation of Components	Visual and Functional
204	Fuel Lines, Hose, Vents, Shut-Off Valve	Improper Application	Visual
205	Painting	Improper Application	Visual
206	Workmanship	Not Following Good Practice, Improper Installation or Adjustments of Components	Visual and Functional

4.5.2.1 Unclassified characteristics. All defects having no bearing on function, safety, interchangeability or life, but which are considered departure from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots or zero lots within a 30-day period, will be added to the minor classification of defects of the specification with no increase in AQL's.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking, shall be in accordance with the applicable packaging Data Sheet as furnished at the time of invitation for bids; for the level of preservation and packaging, and level of packing specified (see 6.2).

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6. NOTES

6.1 Intended use. The engines covered by this specification are intended for production use, as spares, or as replacements for military tactical and transport vehicles.

6.2 Ordering data. Procurement documeents should specify thhee following

(a) Title, number, and date of this specification.
(b) Drawing number (see 3.2)
(c) Address to which first production sample shall be sent (see 3.1).
(d) Form and number of copies of log sheet to be furnished for production runs and acceptance tests (see 3.7).
(e) Additional data for nameplate, when required (see 3.9.1).
(f) Selection of method and applicable level of preservation and packaging, and level of packing of the referenced specification (see 5.1).

6.3 Bare engine. The term bare engine (see 3.5.3, 3.5.4, and 3.5.6) is defined as an engine depicted on Ord. Detroit Diesel Engine No. 5103048.

6.4 Referee grade oil. The following designations apply for the various grades (see 3.4.9).

Grade	Government Designation	Mfg. Designation
10	MB2	CHL-140
30	MB1	CHL-139
50	None	None

6.5 Brake horsepower. Brake horsepower is defined as the horsepower output of the bare engine at the flywheel.

6.6 Exhaust smoke density. Exhaust smoke density should be determined with a Robert Bosch Model EFAW 68 Smoke Meter or equal (as approved by responsible engineering activity), and Model DFAW 65 Sampling Pump or equal (as approved by responsible engineering activity). The following table may be used to visually define the degree of exhaust smoke density in lieu of the smoke meter:

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Description of Exhaust Smoke	Classification
Clear	1
Haze	2
Light Gray	3
Medium Gray	4
Dark Gray to Black	5

NOTE: Observation of exhaust smoke should be made against a white background within 3 feet of the turbocharger outlet.

6.7 Electromagnetic compatibility test. The contracting officer should submit a copy of the report(s) required in 4.5.11 to the Commanding General, U.S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey for approval.

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2815-A133

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OMB Approval
No 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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